

Vol 12 Chapter 6 — Spectroscopy as Substrate Cartography

Keeper (author pass — deep math/physics revision)

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Chapter 6 — Spectroscopy as Substrate Cartography

Extended-reach chapter (May 2026), identification tier. Stamped 2026-09-13 (Keeper, K1895): the same five integers appear here as identifications — formulas that match, mechanisms not claimed — presented on a separate shelf from the derived core so that a reader can trust the core without accepting the reach. Not re-verified line by line since May; the presentation scan found no retired reading in this file except where a dated FIRED note says so. Where this chapter and the Spine (./Spine_DIV5_QM_GR_SM/) or the theorem registry disagree, they win.

Level 1 — one sentence

Spectroscopy — UV/Vis, IR, NMR, microwave, X-ray, mass — measures atomic and molecular substrate K-type transitions at different energy scales, each technique mapping a specific substrate K-type spectrum under specific probe boundary conditions; in BST, spectroscopy is substrate cartography of the substrate's natural K-type structure.

Level 2 — graduate-physicist precision

6.1 Spectroscopic techniques

Technique	Energy / wavelength	Probes
UV/Vis	1-10 eV / 100-700 nm	Electronic transitions
IR	0.01-1 eV / 1-100 μm	Vibrational
NMR	μeV / radio frequency	Nuclear spin
Microwave	meV / mm-cm	Rotational
X-ray	keV / 0.01-10 nm	Core-electron
Mass spec	molecular mass	Identification

6.2 Selection rules

Allowed transitions follow selection rules: - Electric dipole (E1): $\Delta\ell = \pm 1$, $\Delta j = 0, \pm 1$, $\Delta m = 0, \pm 1$ - Magnetic dipole (M1), electric quadrupole (E2): for forbidden E1 - Vibrational: $\Delta v = \pm 1$ (harmonic; overtones from anharmonicity) - Rotational: $\Delta J = \pm 1$

Substrate reading: selection rules are substrate K-type angular momentum + parity conservation.

6.3 NMR

Nuclear spins in magnetic field absorb at Larmor frequency $\omega_L = \gamma B$. Chemical shift δ in ppm; couplings J in Hz.

Used for: molecular structure determination (organic chemistry), protein structure (with isotopic labeling), MRI medical imaging.

6.4 Mass spec

Ionize, accelerate, separate by m/z . Identify molecular fragments. Combined with chromatography (GC-MS, LC-MS) for complex mixtures.

6.5 BST substrate eigentone predictions

Beyond standard spectroscopy: BST predicts specific substrate eigentones in BST-engineered configurations: - BaTiO₃ 137-plane (Vol 9 Ch 8) - Photonic crystal (Vol 9 Ch 9) - Possibly others (substrate-anchored materials, BST primary integer ratios)

These would manifest as anomalous absorption / emission features at substrate-predicted frequencies.

6.6 K-audit anchors

- Vol 5 Ch 3 (selection rules from angular momentum)
- Vol 9 Ch 8, 9 (BST substrate eigentone tests)

Level 3 — 5th-grader accessibility

Spectroscopy = shine light/microwaves/X-rays/etc. at a sample, see what frequencies it absorbs. Each technique probes different molecular structure: - UV/Vis: electronic transitions (colors of dyes) - IR: vibrations (chemical bonds) - NMR: nuclear spins (organic chemistry structure, MRI) - Mass spec: molecular masses - X-ray: core electrons + crystal structure

Selection rules govern allowed transitions. In BST: spectroscopy is substrate cartography — mapping the substrate's K-type structure. BST eigentone predictions (Vol 9): BaTiO₃ 137-plane + photonic crystal predicted to show substrate-natural resonances (\$25K + \$10K experiments).

What comes next

Chapter 7 develops crystallography (BST_Vol12 chapter from earlier session).

Where to look this up

- Hollas, *Modern Spectroscopy*
- BST: Vol 5 Ch 3; Vol 9 Ch 8, 9